

Shifts in monsoon transition periods intensify air pollution exposure in Seoul, South Korea

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Plain Language Summary

This study examined how air pollution and rainfall patterns in Seoul, South Korea have changed over the past 35 years. We found that the summer monsoon season – when heavy rains help clean pollution from the air – now arrives one month later than it did in the 1990s and early 2000s [1]. More importantly, we discovered that the worst air pollution from fine particles (PM_{2.5}) has shifted dramatically from springtime to winter, occurring three months earlier than before. This change has led to a 50% increase in people's exposure to harmful fine particles during the months between the worst and best air quality. Winter is already a challenging time for respiratory health due to the cold and flu season, so this shift means Seoul's nearly 10 million residents now face their highest pollution exposure when they're most vulnerable to breathing problems [2,3]. These changes require immediate attention from policymakers to protect public health, including updating air quality warnings, improving winter heating systems to reduce emissions, and working with neighboring countries to address cross-border pollution. As climate change continues to affect weather patterns, cities like Seoul must adapt their air quality management strategies to protect residents from evolving pollution threats.

1. Introduction

Air pollution represents one of the most pressing public health challenges of the 21st century, with the World Health Organization estimating that outdoor air pollution contributes to approximately 4.2 million premature deaths annually worldwide [4]. Urban areas in East Asia face particularly severe air quality challenges, with megacities like Seoul, Beijing, and Shanghai regularly experiencing particulate matter concentrations far exceeding WHO Air Quality Guidelines [5,6]. Fine particulate matter (PM_{2.5}) – particles smaller than 2.5 micrometers in diameter – poses especially grave health risks, penetrating deep into the lungs and entering the bloodstream, where it contributes to cardiovascular disease, respiratory illness, stroke, and lung cancer [7]. Seoul, South Korea's capital and largest city with a metropolitan population of approximately 25 million, has struggled with persistent air quality challenges despite significant regulatory efforts and emission controls implemented over recent decades [3,8].

The seasonal dynamics of air pollution in East Asia are intimately linked to the region's monsoon climate system [9]. The East Asian summer monsoon, characterized by heavy rainfall from June through August, provides a critical atmospheric cleansing mechanism through wet deposition, wherein precipitation removes pollutants from the atmosphere [10,1]. This seasonal pattern has historically resulted in predictable cycles of air quality, with pollution peaks occurring during dry winter and spring months and troughs during the summer monsoon season. However, emerging evidence suggests that climate change is altering monsoon timing, intensity, and duration across

Asia, with potentially significant implications for air quality dynamics and public health [11].

Understanding temporal shifts in both meteorological patterns and pollution seasonality is crucial for effective air quality management and public health protection [12]. If seasonal pollution patterns are changing – either due to shifting emission sources, evolving meteorological conditions, or altered regional transport pathways – then air quality management strategies, public health advisories, and policy interventions based on historical patterns may no longer be effective or appropriately timed. Moreover, changes in the timing of pollution peaks relative to seasonal health vulnerabilities (such as respiratory infection season) could substantially alter population health burdens beyond what concentration levels alone would suggest.

This study addresses these critical knowledge gaps by conducting a comprehensive temporal analysis of air quality and precipitation patterns in Seoul from 1988 to 2024, examining whether and how seasonal patterns have shifted over this nearly four-decade period. We focus particular attention on fine particulate matter (PM_{2.5} and PM₁₀) due to their severe health impacts and their sensitivity to both local emission sources and regional meteorological patterns. By quantifying temporal shifts in peak and trough timing, assessing changes in cumulative pollutant exposure, and linking these changes to evolving precipitation patterns, this research provides essential insights for adaptive air quality management in Seoul and similar East Asian megacities facing the dual challenges of persistent pollution and climate change. The findings have immediate relevance for policy development, public health protection, and regional cooperation strategies

aimed at safeguarding the health and wellbeing of millions of urban residents in one of the world's most economically and demographically significant regions.

2. Data

2.1. Air Quality & Pollutants

The Seoul air quality data in this study was obtained from measurements of several key air pollutants collected and reported by 서울 열린데이터 광장 (Seoul Open Data Plaza). This publicly available dataset provides comprehensive air quality monitoring data from stations across Seoul, enabling a detailed analysis of pollutant concentrations and their relationship with meteorological factors. The following subsections describe each pollutant examined in this study.

2.1.1 PM_{2.5}

Particulate Matter 2.5 (PM_{2.5}) refers to fine inhalable particles with diameters of 2.5 micrometers or less. These particles are particularly concerning for public health as they can penetrate deep into the lungs and even enter the bloodstream [13]. PM_{2.5} in urban environments primarily originates from vehicle emissions, industrial processes, construction activities, and long-range transport from neighboring regions [14]. In Seoul, PM_{2.5} levels are influenced by both local emission sources and transboundary pollution from China and other regions [15]. The World Health Organization (WHO) recommends an annual mean concentration of 5 µg/m³ for PM_{2.5} in their Air Quality Guideline for countries [5].

2.1.2. PM₁₀

Particulate Matter 10 (PM₁₀) consists of inhalable particles with diameters of 10 micrometers or smaller. While larger than PM_{2.5}, PM₁₀ particles can still be inhaled and cause respiratory problems. Common sources include dust from roads, construction sites, industrial emissions, and natural sources such as pollen and soil erosion [14]. PM₁₀ concentrations in Seoul exhibit seasonal variations, with higher levels typically observed during spring months due to Asian dust events (yellow dust or "hwangsa") [16]. The WHO recommends an annual mean concentration of 15 µg/m³ for PM₁₀ [5].

2.1.3. SO₂

Sulfur dioxide (SO₂) is a colorless gas with a pungent odor, primarily produced by the combustion of fossil fuels containing sulfur, such as coal and oil. Major sources include power plants, industrial facilities, and diesel vehicles [14]. SO₂ is a precursor to acid rain and can contribute to the formation of secondary particulate matter [17]. Exposure to SO₂ can cause

respiratory irritation and exacerbate existing cardiovascular and pulmonary conditions [18]. In recent decades, SO₂ levels in Seoul have declined significantly due to the implementation of stricter emission controls and fuel quality standards [19].

2.1.4. NO₂

Nitrogen dioxide (NO₂) is a reddish-brown gas formed during high-temperature combustion processes. The primary source of NO₂ in urban areas is vehicular traffic, particularly diesel engines, along with power generation and industrial activities [14]. NO₂ contributes to the formation of ground-level ozone and secondary particulate matter, and can cause inflammation of airways and reduced lung function [20,21]. Seoul's NO₂ concentrations are closely linked to traffic patterns, with higher levels observed during rush hours and in areas with dense vehicular activity [22].

2.2. Precipitation

Precipitation data for this study was obtained from National Oceanic and Atmospheric Administration's National Centers for Environmental Information. The dataset includes daily precipitation measurements recorded in centimeters (cm) for the study period 1988 to 2024. Precipitation plays a crucial role in air quality dynamics through wet deposition, which removes pollutants from the atmosphere [1]. Rain and snow can effectively "wash out" particulate matter and gaseous pollutants, leading to improved air quality during and after precipitation events [23]. Understanding the relationship between precipitation patterns and pollutant concentrations is essential for comprehending air quality variations in Seoul, particularly given the city's distinct seasonal precipitation patterns with a monsoon season during summer months [24].

3. Methodology

3.1. Study Area and Time Period

This study examines air quality and precipitation data in Seoul, South Korea, a metropolitan area in the mid-latitudes of the Northern hemisphere. The area is impacted by Westerlies in the winter and Southerlies in the summer during the monsoon season.

The specific time period studied is 1988 to 2024, which follows South Korea's era of rapid industrialization from the 1960s to 1980s [25]. This period was also selected for the lack of availability of detailed data prior to this time period. For temporal comparison, the data was also split into an early and recent period, with the specific years dependent on the time range of the measurements.

3.2. Data Quality Control

3.2.1. Normalization

Before applying normalization, outliers were removed using the 1.5 interquartile range (IQR) rule, in which measurements outside 1.5x the length of the IQR from the first or third quartile are removed. Then, min-max normalization was applied, making all values fall between the interval [0, 1] by using $(\text{value} - \text{min}) / (\text{max} - \text{min})$. This specific method of normalization was selected to enable comparison across pollutants with different scales.

Different measures of central tendency (value) were selected for precipitation and air quality data based on the characteristics of their distributions. For air pollutants, monthly median concentrations were used because they are less influenced by outliers and extreme pollution episodes, providing a more representative measure of typical exposure levels that is commonly used in air quality epidemiology and policy applications. For precipitation, monthly means were used because median values consistently returned zero in many months due to the sporadic nature of rainfall events in Seoul, where more than half of the days in a month may have no precipitation. This resulted in uninformative median values that failed to capture seasonal precipitation patterns.

3.3. Temporal Analysis Methods

3.3.1. Seasonal Pattern Analysis

To characterize the seasonal variability of air pollutants and precipitation, monthly concentration patterns were analyzed by aggregating daily measurements into monthly means (for precipitation) or medians (for air pollutants). The study period was divided into two equal segments: an early period and a recent period, enabling comparison of temporal shifts in seasonal patterns. For each period, peak months (highest concentrations) and trough months (lowest concentrations) were identified for each pollutant. Peak months typically represent periods of maximum pollutant accumulation due to meteorological conditions and emission sources, while trough months often correspond to periods of enhanced dispersion or wet deposition, particularly during the monsoon season. Seasonal patterns were visualized using smoothed curves generated through cubic spline interpolation to better illustrate trends across the calendar year.

3.3.2. Trend Analysis

Long-term trends in pollutant concentrations and precipitation were assessed using linear regression analysis applied to annual aggregated data. The slope of the regression line indicates the rate of change per year, with units corresponding to the pollutant concentration

scale ($\mu\text{g}/\text{m}^3/\text{year}$ for $\text{PM}_{2.5}$ and PM_{10}) or precipitation (cm/year).

To assess the statistical significance and direction of monotonic trends without assuming linearity, Spearman's rank correlation coefficient was calculated between year and concentration values. This non-parametric measure is robust to outliers and appropriate for detecting consistent upward or downward trends over time. Additionally, peak timing shift analysis was conducted to determine whether the month of maximum pollutant concentration has changed systematically over the study period. For each year, the month with the highest mean concentration was identified, and Spearman correlation was used to test whether peak timing has shifted earlier or later across years.

3.4. Exposure Assessment

3.4.1. Area Under the Curve (AUC) Analysis

To quantify cumulative pollutant exposure during the transition from peak to trough concentrations, the area under the curve (AUC) was calculated for the months spanning from the peak month to the trough month in each period. Using the trapezoidal rule for numerical integration, the area under the monthly concentration curve was approximated by summing the areas of trapezoids formed between consecutive monthly data points. Mathematically, for monthly concentrations $C_1, C_2, \dots, C_n$ at months $m_1, m_2, \dots, m_n$:

$$\text{AUC} = \sum [(C_i + C_{i+1}) / 2] \times (m_{i+1} - m_i)$$

The AUC provides a measure of cumulative exposure that integrates both the magnitude and duration of elevated pollutant levels. A larger AUC indicates greater total exposure during the peak-to-trough transition period. To compare exposure between the early and recent periods, percentage change was calculated as:

$$\text{Percentage Change} = [(AUC_{\text{recent}} - AUC_{\text{early}}) / AUC_{\text{early}}] \times 100\%$$

Negative values indicate reduced cumulative exposure (improved air quality), while positive values indicate increased exposure (degraded air quality) during the recent period compared to the early period.

3.5. Statistical Significance Testing

3.5.1. Mann-Whitney U Test

The Mann-Whitney U test, a non-parametric alternative to the independent samples t-test, was employed to compare daily pollutant concentrations between the early and recent periods. This test assesses whether two independent samples are drawn from distributions with the same median, without assuming

normality. It is particularly appropriate for air quality data, which often exhibit skewed distributions. The test was applied to all daily concentration values within the peak-to-trough month range for each period. Statistical significance was evaluated at $\alpha = 0.05$, with p-values less than 0.05 indicating that the distributions of pollutant concentrations differ significantly between the two time periods. The Mann-Whitney U test is robust to outliers and makes no assumptions about the shape of the underlying distributions.

3.5.2. Paired/Wilcoxon Tests

To test whether monthly mean concentrations changed significantly between the early and recent periods, paired statistical tests were conducted. First, the Shapiro-Wilk test was applied to assess normality of the monthly mean differences. If both early and recent monthly means exhibited normal distributions ($p > 0.05$ for Shapiro-Wilk test), a paired t-test was used to compare matched months between periods. If normality assumptions were violated, the non-parametric Wilcoxon signed-rank test was employed instead. These paired tests account for the dependency structure between the same calendar months across different time periods, providing more statistical power than independent samples tests. The null hypothesis states that the median difference between paired observations is zero, with rejection at $\alpha = 0.05$ indicating significant temporal changes in monthly concentrations.

3.5.3. Bootstrap Confidence Intervals

To provide robust estimates of uncertainty in AUC differences and avoid parametric assumptions, bootstrap resampling was performed with 10,000 iterations. In each iteration, years were randomly sampled with replacement from both the early and recent periods, preserving the temporal structure within each year. For each resampled dataset, monthly means were recalculated, and the AUC was computed for the peak-to-trough range. The distribution of AUC differences ($AUC_{\text{recent}} - AUC_{\text{early}}$) across all bootstrap iterations was used to construct 95% confidence intervals using the percentile method (2.5th and 97.5th percentiles). If the confidence interval does not include zero, the change in cumulative exposure is considered statistically significant. Bootstrap methods are particularly valuable for complex statistics like AUC, where analytical standard errors are difficult to derive, and they make minimal distributional assumptions.

3.5.4. Month-by-Month Comparisons

To identify specific months exhibiting significant changes in pollutant concentrations, Mann-Whitney U tests were conducted separately for each of the 12 calendar months, comparing daily values from the early

period to those from the recent period. This approach reveals whether changes are uniformly distributed across the year or concentrated in particular seasons.

Given that 12 tests were performed, consideration was given to the increased risk of Type I errors (false positives) due to multiple comparisons. Months with p-values below 0.05 were flagged as showing significant changes, and patterns across months were examined to identify seasonal shifts in air quality. This granular analysis helps determine whether observed trends are driven by changes throughout the entire year or by shifts in specific seasons, such as winter heating periods or summer monsoon effects.

3.6. Visualization Methods

3.6.1. Seasonal Pattern Plots

Seasonal concentration patterns were visualized using dual-panel figures showing early and recent periods side-by-side. To create smooth, visually interpretable curves from discrete monthly data points, cubic spline interpolation was applied with 300 interpolation points across the 12-month cycle. Peak and trough months were highlighted with vertical dashed and dotted lines, respectively, color-coded by pollutant. Shaded regions (red for peaks, green for troughs) were added to emphasize the timing of extreme concentrations. When peak or trough timing shifted between periods, arrows were overlaid on the recent period panel to indicate the direction and magnitude of the shift, with annotated text boxes displaying the number of months shifted. These visualizations enable direct visual comparison of seasonal patterns and facilitate identification of temporal shifts in pollution cycles.

3.6.2. Time Series Plots

Long-term trends were illustrated through multiple time series visualizations. Annual mean or median concentrations were plotted against year, with linear regression trend lines superimposed to show the direction and rate of change. The slope of the trend line was displayed in the legend for quantitative interpretation.

3.7. Software and Tools

Programming language: Python 3.9.13

Key libraries:

- pandas: Data manipulation
- numpy: Numerical computations
- matplotlib/seaborn: Visualization
- scipy: Statistical tests and interpolation
- scipy.integrate: AUC calculations

3.8. Limitations

Several limitations should be acknowledged in this study. First, data gaps and missing values were present in both air quality and precipitation datasets, particularly in earlier years when monitoring networks were less developed. Missing data were excluded from analyses rather than imputed, which may introduce bias if missingness is non-random.

Second, spatial coverage is limited to monitoring stations within Seoul, which may not fully represent air quality in areas with fewer monitors or in surrounding regions. Concentration gradients within the city due to localized emission sources (e.g., traffic corridors, industrial zones) are not captured in city-wide averages.

Third, the temporal resolution of daily data aggregated to monthly means or medians may obscure short-term pollution episodes or diurnal patterns that could be relevant for acute health effects. Fourth, outlier removal using the IQR method, while reducing the influence of extreme values and potential measurement errors, may also exclude legitimate extreme pollution events, potentially underestimating the full range of exposure.

Finally, statistical tests assume independence of observations and stationarity of underlying processes, which may not fully hold for autocorrelated time series data. Despite these limitations, the methods employed provide robust insights into long-term trends and seasonal patterns in air quality and precipitation in Seoul.

4. Results

4.1. Seasonality of Pollutants & Monsoons

4.1.1 Air Pollutants

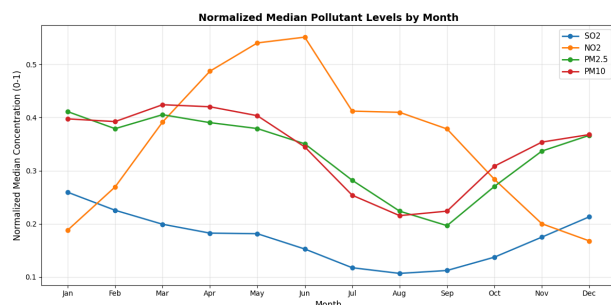


Figure 1. Normalized median pollutant levels of SO₂ (blue), NO₂ (orange), PM_{2.5} (green), and PM₁₀ (red) by month. Monthly patterns of normalized median concentrations for four major air pollutants showing winter peaks for NO₂ and particulate matter, with summer minima corresponding to monsoon season.

For all the air pollutants, there is significant ($p < 0.05$) seasonality. PM_{2.5}, PM₁₀, and SO₂ follow similar patterns – peaking in late winter or spring with a low in the fall. NO₂ differs, however, with a peak in early

summer and a trough in the winter. The seasonality found in PM_{2.5}, PM₁₀, and SO₂ coincides with the typical monsoon season in South Korea.

4.1.2. Precipitation

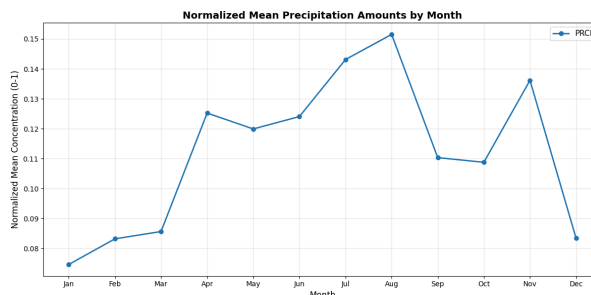


Figure 2. Normalized Mean Precipitation Amounts by Month. Monthly distribution of normalized mean precipitation showing the characteristic East Asian monsoon pattern with peak rainfall during June – August.

Precipitation data showed significant seasonality ($p < 0.05$), with peak rainfall occurring in July during Seoul's monsoon season. Summer months (June–August) accounted for the majority of annual precipitation, while winter and spring months (December–April) exhibited minimal rainfall. This pronounced seasonal pattern corresponds to the East Asian monsoon regime and represents a critical mechanism for atmospheric cleansing of air pollutants, contributing to the observed summer troughs in particulate matter concentrations.

4.2. Precipitation Trend Over Time

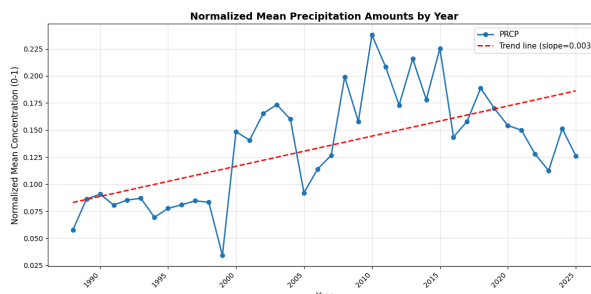


Figure 3. Normalized Mean Precipitation Amounts by Year. Annual trend in normalized mean precipitation from 1988 to 2024 with linear regression trend line (slope = 0.003).

Long-term analysis of precipitation trends revealed a statistically significant ($p < 0.05$) slight increasing trend over the 36-year study period, with a normalized slope of 0.003 per year. This indicates that annual precipitation increased by approximately 0.003 units on the 0-1 normalized scale each year, corresponding to a 3% increase in the normalized range per decade.

While the trend is positive, the relatively small slope suggests modest year-to-year changes in total annual precipitation. Substantial inter-annual variability was observed, with notable peaks and troughs throughout the time series, reflecting the influence of large-scale climate phenomena such as El Niño-Southern Oscillation (ENSO) and natural variability in the East Asian monsoon system. Despite this variability, the overall positive trend may have implications for wet deposition effects on air quality over the study period.

4.3. Longer Monsoon Transition Periods

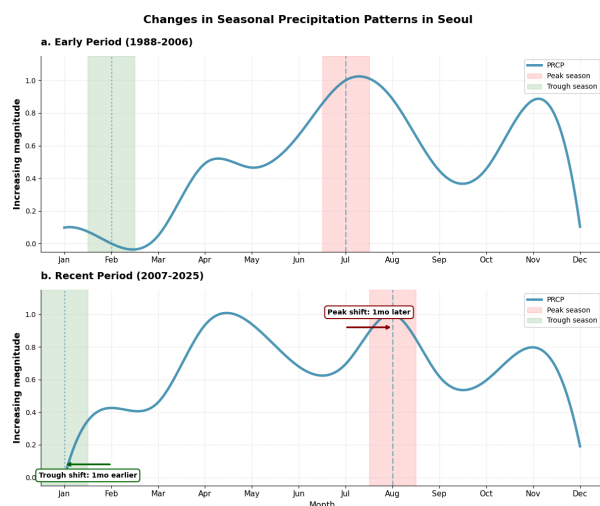


Figure 4. Changes in Seasonal Precipitation Patterns in Seoul. Normalized seasonal precipitation patterns comparing (a) Early Period (1988-2006) and (b) Recent Period (2007-2024), showing shifts in peak timing (red shading) and trough timing (green shading) between periods.

Comparison of seasonal precipitation patterns between early and recent periods revealed notable shifts in monsoon timing. In the early period (1988-2006), peak precipitation occurred in July, while in the recent period (2007-2024), the peak shifted to August, representing a 1-month later shift. Similarly, the trough month (period of minimum precipitation) shifted from February in the early period to January in the recent period, a 1-month earlier shift.

These temporal shifts suggest alterations in the East Asian monsoon onset and duration, with implications for seasonal air quality patterns. The delayed peak precipitation timing may affect the duration and intensity of wet deposition processes that cleanse air pollutants, potentially influencing the observed changes in PM_{2.5} and PM₁₀ seasonal patterns. The shift toward later summer peak rainfall could

extend the period of pollutant accumulation in spring and early summer months.

4.4. Effects of Longer Monsoon Transition Periods

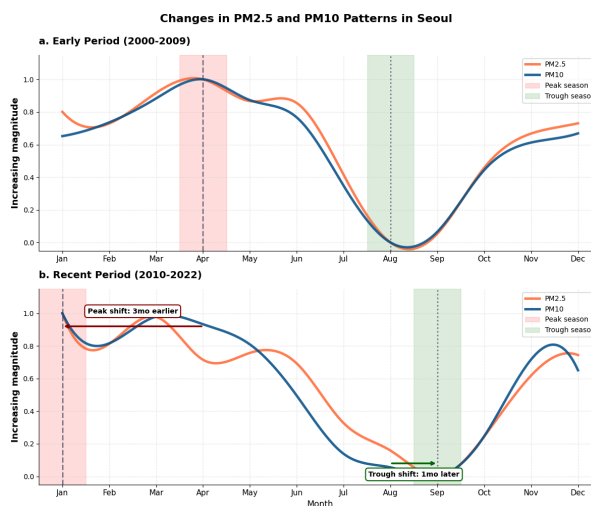


Figure 5. Normalized seasonal patterns of PM_{2.5} (orange) and PM₁₀ (blue) for (a) Early Period (2000-2009) and (b) Recent Period (2010-2023). Red shading indicates peak months, green indicates trough months, with arrows showing temporal shifts between periods.

Analysis of temporal shifts in seasonal patterns focused specifically on PM_{2.5} and PM₁₀, as preliminary Spearman correlation analysis revealed that SO₂ and NO₂ did not exhibit statistically significant shifts in peak timing over the study period (SO₂: $\rho = -0.315$, $p = 0.177$; NO₂: $\rho = -0.009$, $p = 0.971$). The absence of significant temporal shifts for these gaseous pollutants suggests that their seasonal patterns have remained relatively stable, likely reflecting consistent seasonal emission sources (e.g., heating in winter) and atmospheric chemistry. In contrast, particulate matter showed pronounced and significant timing shifts, warranting detailed analysis.

Temporal comparison of particulate matter seasonality revealed significant shifts in both peak and trough timing between the early and recent periods. For PM_{2.5}, the peak concentration month shifted from April in the early period to January in the recent period, representing a 3-month earlier shift. The trough month shifted from August to September, indicating a 1-month later shift. PM₁₀ exhibited nearly identical patterns, with peak shifting from April to January (3 months earlier) and trough shifting from August to September (1 month later).

These shifts resulted in substantial changes in cumulative pollutant exposure: area under the curve (AUC) analysis from peak to trough showed a 50.7% increase in PM_{2.5} exposure ($AUC_{\text{early}} = 117.66$, AUC_{recent}

= 177.34, $p < 0.001$), indicating significantly worsened air quality during the peak-to-trough transition period. In contrast, PM_{10} showed a modest 1.9% reduction in cumulative exposure ($AUC_{early} = 226.93$, $AUC_{recent} = 222.59$), though this change was not statistically significant (95% CI: [-23.42, 14.06]). Mann-Whitney U tests confirmed that the distributions of daily concentrations differed highly significantly between periods for both pollutants ($PM_{2.5}$: $p < 0.001$, $n_{early} = 1,440$, $n_{recent} = 2,687$; PM_{10} : $p < 0.001$, $n_{early} = 1,464$, $n_{recent} = 2,120$). Bootstrap confidence intervals for the $PM_{2.5}$ AUC difference [47.89, 71.35] confirmed the significant increase in cumulative exposure.

The concordance between $PM_{2.5}$ and PM_{10} timing shifts, particularly the 3-month earlier peak occurrence, suggests common meteorological drivers and potential changes in regional emission patterns or atmospheric circulation that now concentrate pollutants in winter months rather than spring.

5. Discussion

5.1. Temporal Shifts in Precipitation and Monsoon Patterns

This study identified a statistically significant 1-month delay in peak precipitation timing in Seoul, shifting from July during the early period (1988-2006) to August in the recent period (2007-2024). This delayed monsoon peak, coupled with a 1-month earlier trough (from February to January), suggests fundamental changes in the East Asian monsoon system over the past four decades. These findings align with broader climate research indicating alterations in monsoon dynamics across Asia, potentially linked to global climate change and shifts in sea surface temperature patterns in the Western Pacific.

The modest but significant increasing trend in annual precipitation (0.003 units/year on normalized scale, $p < 0.05$) suggests intensification of the hydrological cycle in the region. However, the substantial inter-annual variability observed underscores the influence of large-scale climate oscillations such as ENSO, which modulate monsoon strength and timing. The delay in peak monsoon rainfall has critical implications for atmospheric cleansing mechanisms, as wet deposition – the primary removal process for particulate matter – now occurs later in the summer season.

5.2. Divergent Trends in PM Seasonality

The most striking finding of this study is the dramatic 3-month earlier shift in $PM_{2.5}$ and PM_{10} peak concentrations, from April (spring) in 2000-2009 to January (winter) in 2010-2023. This fundamental restructuring of Seoul's particulate matter seasonality

represents a major change in the city's air quality dynamics and carries significant public health implications.

Several factors may contribute to this shift. First, changes in regional meteorological patterns, including modifications to the East Asian winter monsoon and atmospheric circulation, may enhance wintertime stagnation conditions or increase transboundary transport of pollutants from continental sources during winter months. Second, evolving emission patterns – such as increased residential heating demands due to changing temperature patterns, shifts in industrial activity cycles, or changes in regional biomass burning practices – may have redistributed PM sources toward winter months. Third, the observed delay in monsoon precipitation timing may extend the period of pollutant accumulation in early summer, though this does not directly explain the winter peak emergence.

The 1-month delay in trough timing (from August to September) for both $PM_{2.5}$ and PM_{10} appears consistent with the delayed monsoon peak, suggesting that wet deposition remains effective but now occurs later in the season. However, this delay means that the spring-to-summer transition period – previously characterized by declining PM concentrations in anticipation of monsoon rainfall – now experiences prolonged exposure to elevated particulate matter.

5.3. Differential Impacts: $PM_{2.5}$ vs. PM_{10}

While $PM_{2.5}$ and PM_{10} exhibited identical timing shifts, their cumulative exposure trends diverged substantially. $PM_{2.5}$ showed a highly significant 50.7% increase in AUC from peak to trough ($p < 0.001$, 95% CI: [47.89, 71.35]), indicating dramatically worsened air quality during the seasonal transition. In contrast, PM_{10} exhibited a non-significant 1.9% reduction in cumulative exposure (95% CI: [-23.42, 14.06]).

This divergence suggests different driving mechanisms for fine ($PM_{2.5}$) versus coarse (PM_{10}) particulate matter. $PM_{2.5}$, primarily from combustion sources and secondary atmospheric formation, may be increasingly influenced by wintertime residential heating, vehicular emissions under cold-start conditions, and enhanced secondary aerosol formation in stagnant atmospheric conditions. PM_{10} , which includes larger mechanical particles from dust, construction, and road wear, may be less sensitive to these seasonal combustion sources and more influenced by year-round urban activities that have potentially decreased due to improved road infrastructure and dust control measures.

The substantial increase in $PM_{2.5}$ exposure is particularly concerning from a public health perspective, as fine particles penetrate deep into the respiratory system and are associated with

cardiovascular disease, respiratory illness, and premature mortality. The shift toward winter peaks may also compound health risks, as indoor heating and reduced ventilation during cold months increase indoor PM_{2.5} concentrations.

5.4. Stability of Gaseous Pollutant Patterns

The absence of significant temporal shifts in SO₂ and NO₂ peak timing (SO₂: $\rho = -0.315$, $p = 0.177$; NO₂: $\rho = -0.009$, $p = 0.971$) contrasts sharply with the dramatic changes observed for particulate matter. This stability suggests that gaseous pollutant emissions and atmospheric chemistry have remained relatively consistent despite changing meteorological patterns.

SO₂ levels in Seoul have declined substantially over recent decades due to fuel quality improvements and desulfurization technologies, potentially explaining the lack of seasonal pattern shifts – emissions are now sufficiently low that seasonal variations are less pronounced. NO₂, primarily from traffic emissions, maintains stable seasonal patterns reflecting consistent vehicular activity cycles throughout the year, with modest winter heating contributions from combustion sources.

The divergence between stable gaseous pollutants and shifting particulate matter patterns suggests that PM seasonality is more sensitive to meteorological changes and potentially to evolving regional transport patterns, including transboundary pollution from neighboring countries.

5.5. Public Health Implications

The 3-month earlier shift in PM_{2.5} peaks from spring to winter has profound public health implications for Seoul's 9.7 million residents. Winter peaks coincide with respiratory infection season, potentially creating synergistic health burdens. Vulnerable populations – including children, the elderly, and individuals with pre-existing cardiovascular or respiratory conditions – face extended exposure to elevated PM_{2.5} during months when they are already at increased health risk.

The 50.7% increase in cumulative PM_{2.5} exposure from peak to trough represents a substantial escalation in population health risk. Epidemiological studies have consistently demonstrated that chronic PM_{2.5} exposure increases all-cause mortality, with effects observable at concentrations well below current WHO guidelines. The observed shift suggests that annual health burden calculations based on historical seasonal patterns may underestimate current PM_{2.5} impacts.

Furthermore, the later occurrence of the monsoon cleansing effect (trough shifted to September) means that Seoul residents experience prolonged exposure to elevated particulate levels from winter through late

summer – effectively nine months of above-average PM_{2.5} concentrations. This extended exposure window amplifies chronic health effects and may contribute to increasing rates of respiratory disease and cardiovascular events observed in Seoul.

5.6. Policy Implications and Recommendations

These findings carry urgent implications for air quality management and public health policy in Seoul and similar East Asian megacities. Current air quality management plans should be revised to account for shifted seasonal patterns, with emergency response protocols, public health advisories, and pollution reduction measures (e.g., vehicle restrictions, construction bans) traditionally implemented during spring dust events extended or shifted to winter months when PM_{2.5} peaks now occur, prioritizing winter heating emissions from residential and commercial sources for control measures including promotion of clean heating technologies and energy efficiency improvements.

The dramatic shift in PM_{2.5} seasonality may reflect changing transboundary pollution patterns, necessitating enhanced regional cooperation with China and neighboring countries through joint monitoring networks, coordinated emission reduction strategies, and real-time information sharing systems to address regional-scale pollution transport during critical winter months.

Public health agencies should enhance winter PM_{2.5} advisories and expand vulnerable population protection programs, including indoor air quality improvement initiatives (e.g., air purifier subsidies for vulnerable households), outdoor activity guidelines for schools during high-pollution winter days, and health system preparedness for PM-related illness surges. The 50.7% increase in PM_{2.5} cumulative exposure underscores the urgency of aggressive long-term emission reduction strategies, requiring Seoul to accelerate transitions to electric vehicles, expand public transportation, improve building energy efficiency, and phase out fossil fuel combustion in residential and industrial sectors.

Finally, air quality management strategies must be integrated with climate change adaptation planning, recognizing that future meteorological changes may further alter pollution dynamics, including scenario planning for continued monsoon delays, increased precipitation variability, and evolving atmospheric circulation patterns affecting the East Asian monsoon system.

5.7. Broader Regional Context

Seoul's experience may reflect broader trends affecting East Asian megacities. Similar temporal shifts

in particulate matter seasonality have been reported in Beijing, Shanghai, and other major Chinese cities, suggesting regional-scale drivers including climate change impacts on monsoon systems and evolving industrial and transportation emissions across the region. Comparative studies across East Asian cities could elucidate common mechanisms and inform coordinated regional responses.

The delayed monsoon peak observed in Seoul aligns with recent climate model projections indicating weakening and delayed East Asian summer monsoon onset under future climate scenarios. If this trend continues, the window of effective wet deposition for pollutant removal may further contract, potentially exacerbating air quality challenges across the region.

6. Conclusions

This study provides comprehensive evidence of fundamental shifts in air quality and precipitation seasonality in Seoul over recent decades, with potentially impactful public health and policy implications.

6.1. Key Findings

- Precipitation patterns have shifted significantly, with peak monsoon rainfall delayed by 1 month (from July to August) and annual precipitation showing a modest increasing trend (0.003 units/year, $p < 0.05$) over 1988-2024.
- $PM_{2.5}$ and PM_{10} peak timing has shifted dramatically, moving 3 months earlier from spring (April) to winter (January), while trough timing delayed by 1 month (August to September), consistent with the delayed monsoon cleansing effect.
- Cumulative $PM_{2.5}$ exposure increased 50.7% during the peak-to-trough transition period ($p < 0.001$), indicating substantially worsened air quality, while PM_{10} exposure remained stable (1.9% reduction, not significant).
- Gaseous pollutants (SO_2 and NO_2) showed no significant timing shifts, suggesting stable emission sources and atmospheric chemistry despite changing meteorological patterns.
- Strong statistical evidence from multiple tests (Mann-Whitney U, bootstrap confidence intervals, Spearman correlations) confirms the significance and robustness of observed changes.

6.2. Implications

The observed temporal shifts represent a fundamental restructuring of Seoul's air quality dynamics with serious public health consequences. The emergence of winter $PM_{2.5}$ peaks coinciding with the

respiratory infection season, combined with a 50.7% increase in cumulative exposure, substantially elevates health risks for Seoul's population. The delayed monsoon cleansing effect extends the period of elevated pollution exposure, potentially contributing to increased respiratory and cardiovascular disease burden.

These changes demand urgent policy responses, including revised seasonal air quality management strategies, enhanced regional cooperation to address transboundary pollution, expanded public health protection measures, and accelerated long-term emission reductions. The linkage between precipitation pattern shifts and pollution seasonality underscores the need to integrate air quality management with climate change adaptation planning.

6.3. Study Limitations

Several limitations should be acknowledged in this study. Spatial coverage is limited to monitoring stations within Seoul, which may not fully capture pollution gradients in areas with fewer monitors or unique local emission sources. Missing data, particularly in earlier years, may introduce bias if data gaps are non-random, and the IQR-based outlier removal approach may have excluded legitimate extreme pollution events, potentially underestimating exposure variability. Daily aggregation to monthly means/medians obscures short-term pollution episodes and diurnal patterns relevant for acute health effects.

While temporal associations between precipitation and pollution shifts are identified, definitive causal attribution requires additional analysis of emission inventories, meteorological modeling, and source apportionment studies. Time series autocorrelation and potential non-stationarity may violate independence assumptions of some statistical tests, and the binary division into early and recent periods may obscure more complex non-linear temporal trends occurring within each period.

Despite these limitations, the robust statistical evidence from multiple independent tests (Mann-Whitney U, bootstrap confidence intervals, Spearman correlations) supports the validity of the observed temporal shifts.

6.4. Future Research Directions

Several avenues for future research emerge from this study. Detailed source apportionment analysis using chemical composition data, receptor modeling, and air quality simulations could identify specific emission sources and transport pathways driving the observed seasonal shifts and inform targeted mitigation strategies.

Coupling air quality models with regional climate projections could assess whether observed trends align with expected climate change impacts and project future scenarios under different emission pathways.

Epidemiological studies linking the observed 50.7% PM_{2.5} exposure increase to health outcomes (mortality, hospitalizations, respiratory disease incidence) would quantify population health burdens and support cost-benefit analyses of intervention strategies.

Extending this analysis to other East Asian megacities (Beijing, Shanghai, Tokyo) would reveal whether similar temporal shifts are occurring regionally and identify common versus city-specific drivers.

Finally, continued long-term monitoring with enhanced spatial coverage and finer temporal resolution will enable detection of emerging trends, assessment of policy effectiveness, and early warning of future air quality challenges as climate change continues to alter monsoon dynamics and atmospheric circulation patterns across East Asia.

6.5. Final Remarks

The dramatic temporal shifts in PM_{2.5} seasonality documented in this study – a 3-month earlier peak and 50.7% increase in cumulative exposure – represent one of the most significant air quality changes observed in a major global city in recent decades. These findings underscore the dynamic nature of urban air pollution under changing climate and emission conditions and demonstrate that historical seasonal patterns cannot be assumed to persist.

Effective air quality management in Seoul and similar cities requires adaptive strategies that account for evolving pollution dynamics, integration with climate change planning, and strengthened regional cooperation. The health and wellbeing of millions of urban residents depend on urgent action to address these changing air quality challenges. As climate change continues to alter meteorological patterns and monsoon dynamics, proactive monitoring, research, and policy adaptation will be essential to protect public health and achieve sustainable urban air quality.

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